

ADR-001: Technology Decisions

Status

Draft

Date

10 June 2026

Purpose

Document the technology choices selected for the SecOpsAI Platform.

Technology Decisions

1. API Framework

Technology : FastAPI

Reason:

FastAPI was selected because it provides high performance, supports Python-based machine learning integration, and automatically generates API documentation.

2. Database

Technology: PostgreSQL

Reason:

PostgreSQL provides reliability, scalability, and strong support for structured security data.

3. Authentication

Technology: JWT

Reason:

JWT enables secure stateless authentication between users and the API.

4. Event Streaming

Technology: Apache Kafka

Reason:

Kafka allows efficient streaming and processing of security events.

5. Monitoring

Technology: Prometheus and Grafana

Reason:

Prometheus collects metrics while Grafana provides visualization and monitoring dashboards.

